



Cambridge x Evinova (AstraZeneca)

### Trial Parameters

Fill in what you know. Leave blank to let the agent infer.

Indication

e.g. Multiple Myeloma, NSCLC

Phase

Phase 3

Drug Class

e.g. Bispecific, Proteasome In

Administration ?

Unknown / Infer

Patient Population ?

e.g. Relapsed/Refractory, Nev

### Market Scope

HTA Markets

# PRO COA AI Agent

Evidence-based COA instrument selection for oncology clinical trials

University of Cambridge x Evinova (AstraZeneca)

## Describe your trial

Trial description

novel BCMA-targeted bispecific antibody in patients with relapsed/refractory multiple myeloma (RRMM). The administration involves intravenous step-up dosing during Cycle 1 to mitigate CRS risk, followed by biweekly maintenance. Our target product profile

Generate COA Strategy

- ✓ Step 1: Analyzer — extracting trial context and parameters — Relapsed/Refractory Multiple Myeloma | Phase 3 | 4 assumption(s)
- ✓ Step 2: Knowledge Graph — querying trials, reviews, regulatory rules — 21 instruments | 50 reviews | 0 rules
- ✓ Step 3: Scoring Engine — 100-point regulatory scale + battery optimiser — 21 scored | battery: EQ-5D-5L, EORTC QLQ-MY20, EORTC QLQ-C30
- ✓ Step 4: Reasoner — synthesising evidence + live web search — 9726 chars
- ✓ Step 5: Logging to evaluation dataset — Saved to /logs/

COA Strategy Evidence & Scoring Detail Evaluation Log

What the agent understood about your trial —

⌵

What the agent understood about your trial

click to review and correct

The agent inferred the following parameters from your query. If any are wrong, add the correct information to your description and re-run.

⚠ Administration: Step-up dosing inferred from bispecific antibody drug class and explicit user mention of 'intravenous step-up dosing during Cycle 1 to mitigate CRS risk' — confirms FDA 2021 PRO Guidance inference rule for bispecific agents.

⚠ Core domains: Extended RRMM standard domains (bone pain, physical function, treatment tolerability, fatigue per FDA 2021 Multiple Myeloma PRO guidance) with addition of CRS-related symptoms — critical safety monitoring for bispecific antibodies targeting immune activation.

⚠ HTA markets: User explicitly specified NICE and ICER; added EUnetHTA based on Global footprint and EMA submission intent (FDA + EMA stated in input).

⚠ Trial duration cycles: Not provided in user input; marked as Unknown — recommend protocol specification based on disease progression kinetics and safety assessment window for bispecific dosing.

Parameters extracted:

Indication:

Relapsed/Refractory Multiple Myeloma

Phase: Phase 3

Drug class:

Bispecific Antibody

Population:

Relapsed/Refractory

Footprint:

Global

Administration:

Step-up dosing

Core domains (FDA lookup):

physical function, bone pain, treatment tolerability, fatigue, cytokine release syndrome (CRS) symptoms

http://localhost:8501/

Page 2 of 11

**TPP claims:** Delayed time to worsening of physical function, Reduced disease-related bone pain, Manageable treatment tolerability versus standard of care

## Recommended COA Battery

Built by the domain coverage optimiser — selects highest-scoring non-redundant instrument per domain category, plus HTA-required instruments. Covers 5 of 5 required domains.

**Risk Level guide:**  LOW = no regulatory concerns.  MODERATE = addressable risks (e.g. missing MCID).  HIGH = significant risk requiring active mitigation (e.g. not pre-specified).  CRITICAL = fatal regulatory flaw if not addressed (e.g. recall bias with step-up dosing). Note: HTA-required instruments (EQ-5D) show LOW risk even if scientific score is moderate — they are mandatory inclusions, not optional.

### HRQOL GENERAL

 **EQ-5D-5L** LOW Scientific score: 70/100 | Operational: +10 eCOA

### DISEASE SPECIFIC

 **EORTC QLQ-MY20** LOW Scientific score: 90/100 | Operational: +5 open access

### FATIGUE

 **EORTC QLQ-C30** MODERATE Scientific score: 60/100 | Operational: +10 eCOA | +5 open access

### PAIN

 **EQ-5D** LOW Scientific score: 70/100 | Operational: +10 eCOA

### HEALTH UTILITY

 **EQ-5D-3L** LOW Scientific score: 35/100 |

Operational: +10 eCOA

⚠ **Domain coverage gap:** No instrument found for: cytokine release syndrome (crs) symptoms. The Reasoner has been instructed to search the web for instruments covering these domains.

## What the agent understood about your trial

Administration: Step-up dosing inferred from bispecific antibody drug class and explicit user mention of 'intravenous step-up dosing during Cycle 1 to mitigate CRS risk' — confirms FDA 2021 PRO Guidance inference rule for bispecific agents. **What the expert should verify:** The agent interpreted that step-up dosing is used to mitigate CRS risk during Cycle 1, consistent with all FDA-approved T-cell engaging bispecific antibodies which "use step-up dosing to mitigate serious adverse events, including cytokine release syndrome (CRS) and immune effector cell-associated neurotoxicity syndrome (ICANS)." Alternative interpretation could have been continuous dosing or different step-up schedule, which would change the CRS monitoring strategy and recall period requirements for PRO instruments.

Core domains: Extended RRMM standard domains (bone pain, physical function, treatment tolerability, fatigue per FDA 2021 Multiple Myeloma PRO guidance) with addition of CRS-related symptoms — critical safety monitoring for bispecific antibodies targeting immune activation. **What the expert should verify:** The agent inferred CRS symptoms as a core domain based on bispecific

antibodies causing "release of inflammatory cytokines" with "symptoms range from mild flu-like" manifestations. Alternative could have excluded CRS-specific PRO monitoring if relying only on clinician assessment, which would reduce the battery to conventional MM domains.

HTA markets: User explicitly specified NICE and ICER; added EUnetHTA based on Global footprint and EMA submission intent (FDA + EMA stated in input). **What the expert should verify:** EMA submission was inferred from "FDA and EMA" mention, leading to EUnetHTA inclusion for EU Joint Clinical Assessment requirements under Regulation 2021/2282. Alternative interpretation of EU-wide EMA approval could affect utility measurement requirements.

## Recommended COA Battery

| Instrument     | Domain Role      | Score  | Risk     | Why included                                   |
|----------------|------------------|--------|----------|------------------------------------------------|
| EORTC QLQ-MY20 | Disease Specific | 90/100 | LOW      | MM-specific domains with regulatory precedent  |
| EQ-5D-5L       | HRQoL General    | 70/100 | LOW      | HTA utility requirement, preferred version     |
| EORTC QLQ-C30  | Fatigue          | 60/100 | MODERATE | Core MM domain coverage, regulatory acceptance |

**EORTC QLQ-MY20:** In the A.R.R.O.W. trial, QLQ-MY20 showed significantly delayed time to deterioration for disease symptoms with carfilzomib, directly addressing the TPP claim for "reduced disease-related bone pain." The instrument uses a 7-day recall period ("During the past week") and captures bone pain domains essential to this trial: "Have you had bone aches or pain?", "Have you had pain in your back?", "Have you had pain in your hip?" Main risk: EMA rejected QLQ-MY20 for elotuzumab due to null results and for isatuximab due to "null result, no clinically meaningful difference, open-label bias, missing data." **Action required:** Pre-specify as secondary endpoint with alpha-controlled testing hierarchy and ensure adequate power for clinical meaningful difference detection.

**EQ-5D-5L:** In the A.R.R.O.W. trial, EQ-5D-5L showed significantly delayed time to deterioration for both index score and VAS, supporting the TPP claim for "delayed time to worsening of physical function." "NICE Position statement on use of the EQ-5D-5L value set for England (updated October 2019)" confirms EQ-5D-5L as the preferred utility measure, though "there is no currently approved value set for the 5L in England" and "5L descriptive system is increasingly used in clinical studies." **Action required:** Use DSU EEPRU mapping function to UK 3L value set as per NICE 2022 guidance; commission EQ-5D-5L global translations early in trial preparation.

**EORTC QLQ-C30:** In the ELOQUENT-2 trial, QLQ-C30 showed no significant difference between arms but is "FDA/EMA accepted in MM trials" and "used in multiple approved MM drugs" for fatigue and physical functioning domains. Main risk: EMA rejected QLQ-C30 for elotuzumab and designated as exploratory only for isatuximab due to null results and missing data. **Action required:** Pre-specify fatigue and physical functioning subscales as secondary endpoints with appropriate sample size calculation and missing data handling strategy.

# Key Regulatory Risks and How to Mitigate Them

**Risk:** Single-arm bias in PRO data. **Evidence:** FDA and EMA systematically reject PRO claims from single-arm trials due to "single-arm open-label bias subject to systematic overestimation (response bias)" and inability to "exclude bias from concomitant treatments." **Mitigation for THIS trial:** Ensure Phase 3 design includes active or standard-of-care comparator with pre-specified PRO hypotheses and adequate blinding of PRO assessments where feasible.

**Risk:** Null results pattern in MM bispecific trials. **Evidence:** Systematic pattern of null PRO results in MM: elotuzumab, isatuximab, and multiple other agents showed "null result: no improvement in QoL" or "no clinically meaningful difference." **Mitigation for THIS trial:** Power PRO endpoints for clinically meaningful difference thresholds (MY20: 10 points, EQ-5D-5L: 0.044 index points) with 80% power and account for 20-30% missing data inflation factor based on MM trial experience.

**Risk:** CRS-related PRO measurement gap. **Evidence:** Bispecific antibodies "can potentially lead to severe toxicity associated with T-cell activation, particularly cytokine release syndrome (CRS)" with "symptoms range from mild flu-like" symptoms, but no validated CRS-specific PRO instrument is identified in KG records. **Mitigation for THIS trial:** Implement PRO-CTCAE items for fever, fatigue, and constitutional symptoms with 24-hour recall during step-up dosing period (Cycle 1) as validated by "A 24-hour recall period for PRO-CTCAE items has acceptable measurement properties" for daily monitoring.

# HTA Requirements

**NICE:** NICE's preferred measure of health-related quality of life in adults has been EQ-5D since 2008. Current status: EQ-5D-5L included in battery. Consequence of omission: Without EQ-5D, QALY calculation is impossible for NICE appraisals, as "there is no currently approved value set for the 5L in England" but "NICE requires EQ-5D-5L responses to be mapped onto the long-standing EQ-5D-3L value set for the UK." **Action:** Use NICE DSU EEPRU mapping function from 5L to 3L utility values.

**ICER:** ICER uses utility-based measures for cost-effectiveness analysis in US value assessments. EQ-5D-5L is strongly preferred for QALY calculation. Current status: EQ-5D-5L included. Consequence of omission: US value assessment compromised without preference-based utility measure. **Action:** No additional instrument required.

**EUnetHTA:** "EU HTA Regulation 2021/2282 Joint Clinical Assessments increasingly require standardised PRO instruments for cross-country comparison. EQ-5D-5L required for HTA utility analysis" and "EQ-5D-5L is currently the preferred EuroQol instrument for FDA and EMA oncology drug submissions" under "EU Regulation 2021/2282 (Joint Clinical Assessment)." Current status: EQ-5D-5L included. **Action:** Ensure EQ-5D-5L translation availability for all EU trial countries.

## Implementation Checklist

1. **Assessment Schedule:** EORTC instruments (QLQ-MY20 + C30) every 2 cycles starting Cycle 1 Day 1; EQ-5D-5L every cycle; PRO-CTCAE daily during Cycle 1 step-up



dosing with 24-hour recall, then weekly thereafter.

2. **Estimand Strategy:** Apply Treatment Policy estimand per ICH E9(R1) (2019) to handle intercurrent events including treatment discontinuation, CRS management interventions, and concomitant supportive care without data censoring.
3. **eCOA Setup:** Verify eCOA version availability for all instruments; flag mode equivalence studies needed if transitioning from paper versions used in precedent trials.
4. **Translation Commissioning:** EORTC QLQ-MY20 "has been translated into over 70 language versions" and EQ-5D-3L "has now been translated into more than 170 languages." Commission EQ-5D-5L translations for trial languages early as availability may be limited compared to 3L.
5. **Pre-specification Requirements:** Include all PRO endpoints in statistical analysis plan with multiple testing adjustment; specify MCID thresholds for responder analyses: MY20 domains 9-13 points, EQ-5D-5L index 0.044 points ; define missing data handling strategy anticipating 20-30% missing data in advanced MM population.
6. **Alpha Control Strategy:** Position PRO endpoints in testing hierarchy after primary efficacy endpoint to preserve Type I error; consider gatekeeping strategy if multiple PRO claims are pursued simultaneously.

## Evidence Gaps

**CRS-specific PRO instrument validation:** While bispecific antibodies are known to cause "cytokine release syndrome (CRS): An exaggerated systemic inflammatory response triggered by the effects of T-cell engaging

therapies" with specific symptom patterns, the KG does not contain records of validated CRS-specific PRO instruments beyond general PRO-CTCAE items. Web search confirms PRO-CTCAE utility for CRS monitoring but validation data for CRS-specific items or composite scores requires further investigation.

#### **EQ-5D-5L direct value set availability: NICE**

documentation confirms "there is no currently approved value set for the 5L in England" requiring mapping approaches , but the full global availability of EQ-5D-5L value sets for primary health economic analysis across all study countries needs verification for countries beyond UK mapping requirements.

Citations: [TI-XXX] = trial instrument record from KG (see Evidence & Scoring Detail tab). [RR-XXX] = regulatory review record. [IR-XXX] = instrument reference. [REJ-XXX] = rejection reason from medical review. Web citations appear as [Source Name](#) — click to open. Switch to the **Evidence & Scoring Detail** tab to see all KG records with source links.

---

## Language and translation status — per recommended instrument

**KeyError: 'sufficient'**

Traceback:

```
File "/Users/nathalinechanentia/Document:
    result = func()

File "/Users/nathalinechanentia/Document:
    exec(code, module.__dict__) # noqa:

File "/Users/nathalinechanentia/Document:
    icon = "✅" if status["sufficient"]
```

[Copy](#) [Ask Google](#) [Ask ChatGPT](#)